

# Rohan Fozdar

[rfozdar@knox.edu](mailto:rfozdar@knox.edu) | +1 (646)-749-0816 | [linkedin.com/in/rohan-fozdar/](https://www.linkedin.com/in/rohan-fozdar/)

## EDUCATION

### Knox College

Galesburg, IL

Bachelor of Science in Mathematics & Bachelor of Arts in Data Science

Anticipated June 2027

- **GPA:** 3.89 | Achieved Dean's List (9 semesters), Annual Carr Math Exam Winner, Knox Next Genius Scholar
- **Relevant Coursework:** Probability Theory, Monte Carlo Simulations, Stochastic Processes, Options Theory & Portfolio Optimization, Applied Data Structures, Linear & Logistic Regression, Time Series (GARCH) Models, Risk Management, Neural Networks & Deep Learning, NLP, Algorithm Design & Complexity, Database Systems.

## PROFESSIONAL EXPERIENCE

### Quantitative Analytics Intern

Mumbai, India

Monarch Network Capital Ltd.

November 2025 – January 2026

- Designed a momentum trading strategy on NSE ETFs using time trend derivatives (slope estimates over intervals).
- Backtested 1 year of tick-by-tick data achieving 45% returns & 3.43 Sharpe (BankNifty over 1 year).
- Formulated a moving average strategy for NSE ETFs using CMP against 5/10/20/50/200 MA to indicate daily trends. Backtested 25 years of daily data achieving 3100% returns & 2.86 Sharpe (Nifty over 25 years).

### Mathematics Teaching Assistant

Galesburg, IL

Knox College - Department of Science & Mathematics

March 2024 – present

- Developed class material and led recitation sessions for Calculus I-III, Discrete Math, Linear Algebra, Foundations of Analysis, Methods of Proof, Differential Equations, Mathematical Statistics and Financial Math.
- Graded homework and exams and collaborated with other TA's to host tutoring sessions for the department.

### AI Intern

Ahmedabad, India

MeasureOne - Consumer Data Access & Verification Solutions

June 2023 – August 2023

- Spearheaded a team of interns in an AI-automation initiative by fine-tuning early versions of ChatGPT using prompt-completion pairs to match client-specific data tables and generate specialized parser codes.
- Designed a preprocessing pipeline that normalized heterogeneous transcript formats, engineered structured prompts, and validated outputs against production requirements. Reduced manual engineering workload by 25%.

## PROJECTS (Full portfolio - [rohanfozdar.com](https://rohanfozdar.com))

### ERCOT Grid Stress & Electricity Price Forecaster

- Built an ERCOT electricity spot price forecaster using XGBoost trained on 9 data sources across 2016–2024, achieving 58.8% spike recall vs the day-ahead market's 28.9% and detecting stress events up to 144 hours ahead.
- Extended the architecture with IBM Research's Tiny Time Mixers (TTM), to improve prediction of extreme price variations, by combining TTM's zero-shot temporal reasoning with domain-specific sentiment features.

### AutoSearch

<https://job-finder-aa5m8lsoa-rohanfozdars-projects.vercel.app/>

- Shipped an end-to-end job search platform with a Rest API ingestion pipeline, RAG-style semantic scoring using FAISS vector search and sentence-transformers, and an ATS scoring microservice to match better fit roles.
- Engineered an LLM-based ranking engine using PyMuPDF resume parsing, overlapping text chunking, and cosine similarity matching against a FAISS vector store, and NLP-based job classification across 50+ listings per query.

### Binomial Options Pricer

<https://rohanfozdar.github.io/options-pricer/>

- Developed and deployed a options pricing tool through Binomial branching with live data from YFinance for comparison, providing valuations across all possible strikes and expirations, achieving greater accuracy.
- Integrated automated risk-free rate calculations by pulling US treasury yields based on expiration to improve accuracy over traditional statistical measures.
- Implemented a 3D IV surface visualization allowing users to analyze trends across different strikes & expirations, with a module explaining how shifts in IV trends reflect market sentiment and supply & demand imbalances.

## SKILLS AND INTERESTS

**Skills:** JavaScript, Python, PowerBI, R, SQL, C++, Excel, Matplotlib, NumPy, Pandas, Java, TensorFlow, Tableau, Matlab, PyTorch, FastAPI, LangChain, HuggingFace, Snowflake

**Interests:** Aeroscience, Poker, Mechatronics, Automotive Design, Chess, Sports Betting, Architecture

**Organizations:** Knox Consulting Club (President), Honor Board (Co-chair), Quantitative Investment society